

Q1

```
#include <stdio.h>
int main()
{
    float P, R, T, SI;
    printf("enter principal amount");
    scanf("%f",&P);
    printf("enter rate of interest");
    scanf("%f",&R);
    printf("enter time (years)");
    scanf("%f",&T);
    SI = P*R*T/100;
    printf("your simple interest is=%f", SI);
}
```

```
gdb.exe --interpreter=mi
enter principal amount 1200
enter rate of interest 7
enter time (years) 4
your simple interest is=336.000000
PS E:\intro to prog assignments sem1>
```

Q2:

```
#include <stdio.h>
int main()
{
    printf("enter your marks in PHY, MATH, CHEM, ENG, PE");
    int PHY, MATH, CHEM, ENG, PE;
    scanf("%d %d %d %d %d", &PHY, &MATH, &CHEM, &ENG, &PE);
    int total = PHY + MATH + CHEM + ENG + PE;
    printf("your total marks are=%d\n", total);
    int percent = (total/500)*100;
    printf("your percentage is=%f", (float)total / 5);
}
```



```

gdbengine --pid=Microsoft-MIEngine-Pid-dzkntid.hkc'
enter your marks in PHY, MATH, CHEM, ENG, PE 50 60 70 80 90
your total marks are=350
your percentage is=70.000000
PS E:\intro to prog assignments sem1>

```

Q3

```

#include <stdio.h>
int main()
{
    int gross, basic, housing, bonus, overtime;
    printf("enter your basic salary, housing allowance, bonus, and
overtime income");
    scanf(" %d %d %d %d", &basic, &housing, &bonus, &overtime);
    gross = basic + housing + bonus + overtime;
    printf("your gross salary is=%d", gross);
}

```

```

gdbengine --pid=Microsoft-MIEngine-Pid-dzkntid.hkc'
enter your basic salary, housing allowance, bonus, and overtime income 10000 2000 1000 800
your gross salary is=13800
PS E:\intro to prog assignments sem1>

```

Q4

```

#include <stdio.h>
int main()
{
    float temp_f, temp_c;
    printf("enter temperature in fahrenheit");
    temp_c = (temp_f - 32)*5/9;
    scanf("%f", &temp_f);
    printf("temperature in celsius=%f",temp_c);
}

```

```

gdbengine --pid=Microsoft-MIEngine-Pid-dzkntid.hkc'
gdb.exe' '--interpreter=mi'
enter temperature in fahrenheit 500
temperature in celsius=-17.777779
PS E:\intro to prog assignments sem1>

```


Q5

```
#include <stdio.h>
int main()
{
    int a, b, c;
    printf("enter first number");
    scanf("%d", &a);
    printf("enter second number");
    scanf("%d", &b);
    c = a;
    a = b;
    b = c;
    printf("swapped the numbers first=%d\n" "second=%d", a, b);
}
```

```
gdb.exe' '--interpreter=mi'
enter first number19
enter second number20
swapped the numbers first=20
second=19
PS E:\intro to prog assignments sem1>
```

Q6

```
#include <stdio.h>
int main()
{
    int a, b;
    printf("enter first number");
    scanf("%d", &a);
    printf("enter second number");
    scanf("%d", &b);
    a = a*b;
    b = a/b;
    a = a/b;
    printf("swapped the numbers first=%d\n" "second=%d", a, b);
}
```



```
gdb.exe' '--interpreter=mi'
enter first number14
enter second number15
swapped the numbers first=15
second=14
PS E:\intro to prog assignments sem1>
```

Q7

```
#include <stdio.h>
int main()
{
    float base, height, area;
    printf("enter base and height of triangle");
    scanf("%f %f", &base, &height);
    area = 0.5 * base * height;
    printf("area of triangle=%f", area);
}
```

```
gdb.exe' '--interpreter=mi'
enter base and height of triangle10 20
area of triangle=100.000000
PS E:\intro to prog assignments sem1>
```

Q8

```
#include <stdio.h>
int main()
{
    int t = 31558150, tsec, thr, tmin;
    printf("earth takes %f seconds to for a period of revolution",t);
    thr = t/3600;
    tmin = (t%3600)/60;
    tsec = (t%3600)%60;
    printf("period of revolution is %d hours, %d minutes, and %d seconds",
thr, tmin, tsec);
}
```



```
gdb.exe' '--interpreter=mi'
earth takes 0.000000 seconds to for a period of revolutionperiod of revolution is 8766 hours, 9 minut
es, and 10 seconds
PS E:\intro to prog assignments sem1>
```

Q9

```
#include <stdio.h>
int main()
{
    int hrs, mins, sec, total_sec;
    printf("enter hours, minutes, and seconds in (hh mm ss) format");
    scanf("%d %d %d",&hrs,&mins,&sec);
    total_sec = hrs*3600 + mins*60 + sec;
    printf("time given converted to seconds = %d",total_sec);
}
```

```
enter hours, minutes, and seconds in (hh mm ss) format 10 15 6
time given converted to seconds = 36906
PS E:\intro to prog assignments sem1>
```

Q10

```
#include <stdio.h>
int main()
{
    int M, P, C, E, CM;
    printf("enter marks in Mathematics, Physics, Chemistry (out of 200)
and Entrance exam (out of 100)");
    scanf("%d %d %d %d", &M, &P, &C, &E);
    CM = (M + P + C)/2 + E;
    printf("Cutoff marks for given scores= %d", CM);
}
```

```
PS E:\intro to prog assignments sem1> & "E:\intro to prog assignments sem1\que10.exe"
enter marks in Mathematics, Physics, Chemistry (out of 200) and Entrance exam (out of 100)120 130 140
80
Cutoff marks for given scores= 275
PS E:\intro to prog assignments sem1>
```


Q11

```
#include <stdio.h>
int main()
{
    printf("enter total time in seconds");
    int t, hr, min, sec;
    scanf("%d", &t);
    hr = t/3600;
    min = (t%3600)/60;
    sec = (t%3600)%60;
    printf("time in hh:mm:ss format is %d hours,%d minutes,%d seconds",
hr, min, sec);
}
```

```
enter total time in seconds42000
time in hh:mm:ss format is 11 hours,40 minutes,0 seconds
PS E:\intro to prog assignments sem1> █
```

